Basic Principles of Medicine 1

Module: Foundation to Medicine

Lecture: FTM 28

Fibrous Proteins

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Acknowledgment: Dr. Margit Trotz

Objectives: DLA: Proteoglycans and Glycoproteins

- Before lecture 'Fibrous proteins'

SOM.MK.I.BPM1.1.FTM.3.BCHM.0143	Describe the general structure of proteoglycans. Indicate sugar derivatives used for GAG synthesis.
SOM.MK.I.BPM1.1.FTM.3.BCHM.0144	Discuss the general composition and function of GAGs. Describe the bottle-brush structure of proteoglycans.
SOM.MK.I.BPM1.1.FTM.3.BCHM.0145	Describe the synthesis of proteoglycans, hyaluronic acid and the extracellular assembly of proteoglycan aggregates using link proteins leading to shock absorbing aggregates.
SOM.MK.I.BPM1.1.FTM.3.BCHM.0146	Discuss the functions of hyaluronic acid and heparin.
SOM.MK.I.BPM1.1.FTM.3.BCHM.0147	Differentiate O-linked and N-linked glycoproteins and indicate the amino acids involved.
SOM.MK.I.BPM1.1.FTM.3.BCHM.0148	Discuss blood group antigens as an example for O-linked glycoproteins.
SOM.MK.I.BPM1.1.FTM.3.BCHM.0149	Outline synthesis of N-linked glycoproteins and indicate significance of dolichol-PP and mannose.
SOM.MK.I.BPM1.1.FTM.3.BCHM.0150	Highlight the importance of phosphorylation of mannose (Mannose 6-P marker) in sorting of lysosomal enzymes.
SOM.MK.I.BPM1.1.FTM.3.BCHM.0151	Predict the molecular and clinical effects of I-cell disease
SOM.MK.I.BPM1.1.FTM.3.BCHM.0152	Compare concepts of synthesis of O-linked and N-linked glycoproteins.

Objectives: Fibrous Proteins

SOM.MK.I.BPM1.1.FTM.3.BCHM.0153	Describe the amino acid composition, structure and function of collagen. Discuss changes in collagen structure in vitamin C deficiency (scurvy).
SOM.MK.I.BPM1.1.FTM.3.BCHM.0154	Discuss the synthesis of procollagen in fibroblasts and its release into the ECM.
SOM.MK.I.BPM1.1.FTM.3.BCHM.0155	Describe the role of lysyl oxidase (copper containing) in covalent cross-linking between tropocollagen molecules in maintenance of collagen stability.
SOM.MK.I.BPM1.1.FTM.3.BCHM.0156	Describe the molecular and genetic basis of Marfan syndrome.
SOM.MK.I.BPM1.1.FTM.3.BCHM.0157	Discuss the molecular basis for occurrence of fractures and presence of blue sclerae in osteogenesis imperfecta (OI).
SOM.MK.I.BPM1.1.FTM.3.BCHM.0158	Describe the amino acid composition, structure and function of elastin and desmosine and highlight significance of lysyl oxidase.
SOM.MK.I.BPM1.1.FTM.3.BCHM.0159	Describe the molecular basis of Ehlers-Danlos Syndrome (EDS).
SOM.MK.I.BPM1.1.FTM.3.BCHM.0160	Compare and contrast the effect of deficiency of prolyl hydroxylase and lysyl oxidase. Compare and contrast the effects of vitamin C deficiency and copper deficiency.
SOM.MK.I.BPM1.1.FTM.3.BCHM.0161	Differentiate Ehlers-Danlos Syndrome (EDS), osteogenesis imperfecta (OI) and Marfan syndrome based on molecular defect and clinical features

Recommended Reading

- Lippincott's Biochemistry 9th Edition
🔗 https://sgu.idm.oclc.org/login?url=https://premiumbasicsciences.lwwhealthlibrary.com/book.aspx?bookid=3402; Chapter 4: Fibrous proteins

Lippincott's Illustrated Reviews Biochemistry 9th edition; Chapter 4:
Fibrous Proteins | Lippincott Illustrated Reviews: Biochemistry, 9e | Premium Basic Sciences | Health Library

Elastic fiber during aging and disease Aging Research Reviews 66 (2021); Andrea Heinz 73 pages: https://www.sciencedirect.com/science/article/pii/S1568163721000027

Fibrous Proteins, Proteoglycans and Glycoproteins

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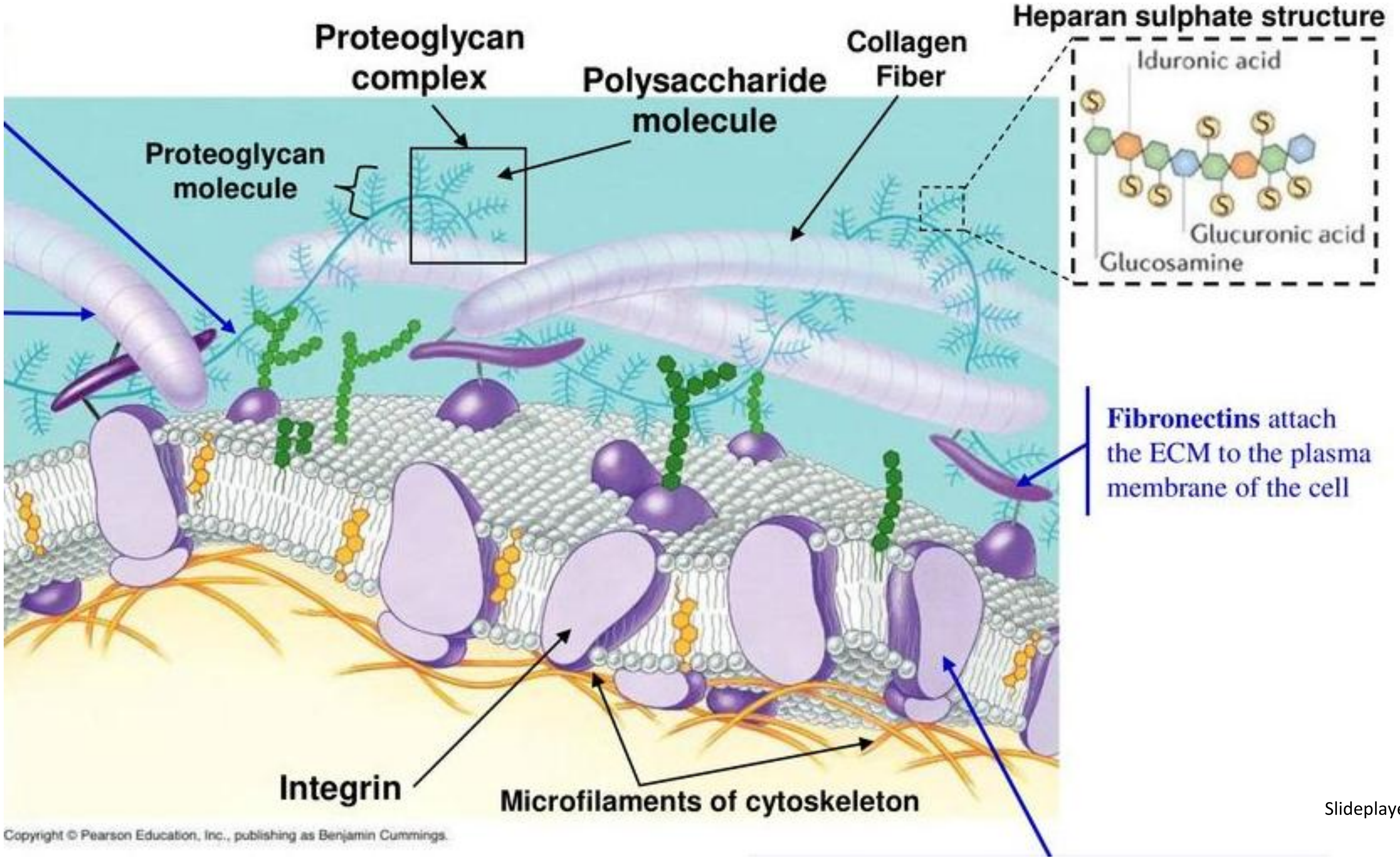
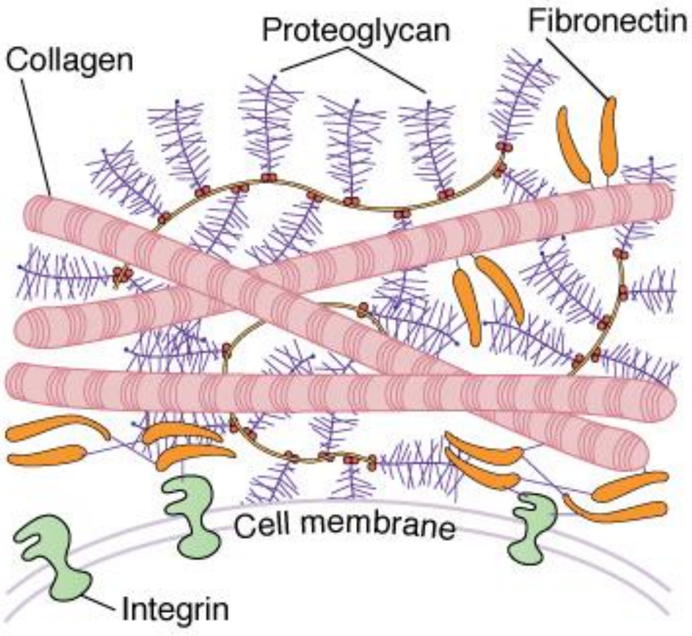
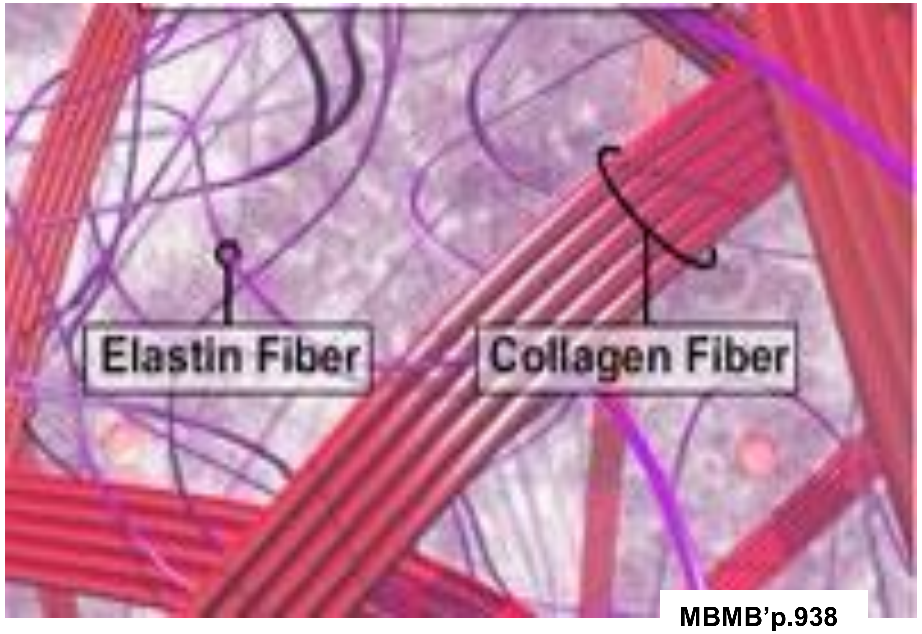
Fibrous proteins and proteoglycans are part of the extracellular matrix (ECM)



The ECM is continuously remodeled and changed



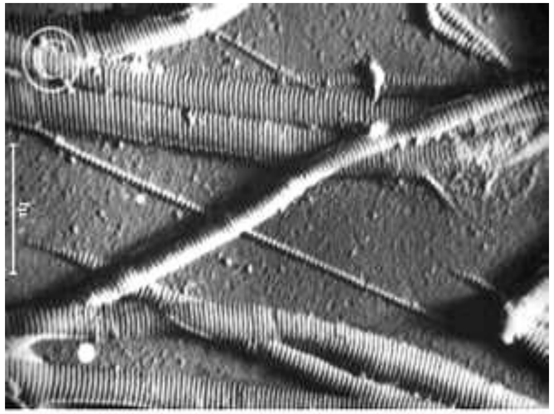


Proteoglycans fill the extracellular space and attract **water.** They are involved in many functions, like shock absorbing in joints and they also facilitate cell migration and wound healing.

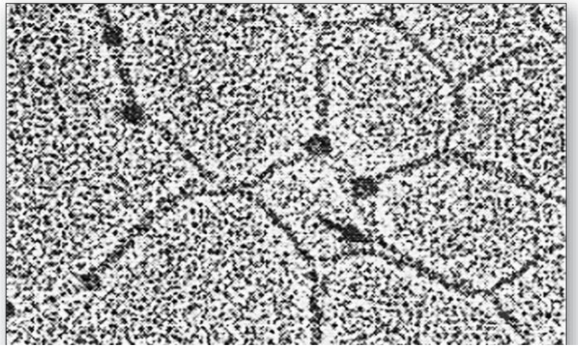
Fibrous proteins include **collagen fibers** and **elastic fibers** which provide **structural support** of surrounding cells. Elastic fibers also interact with other matrix molecules involved with cell signaling and modulation of growth factors

The functions of collagen include forming of fibrils and of networks

	Fibril-forming
I	Skin, bone, tendon, blood vessels, cornea
II	Cartilage, intervertebral disk, vitreous body
III	Blood vessels, fetal skin



	Network-forming
IV	Basement membrane
VII	Beneath stratified squamous epithelia



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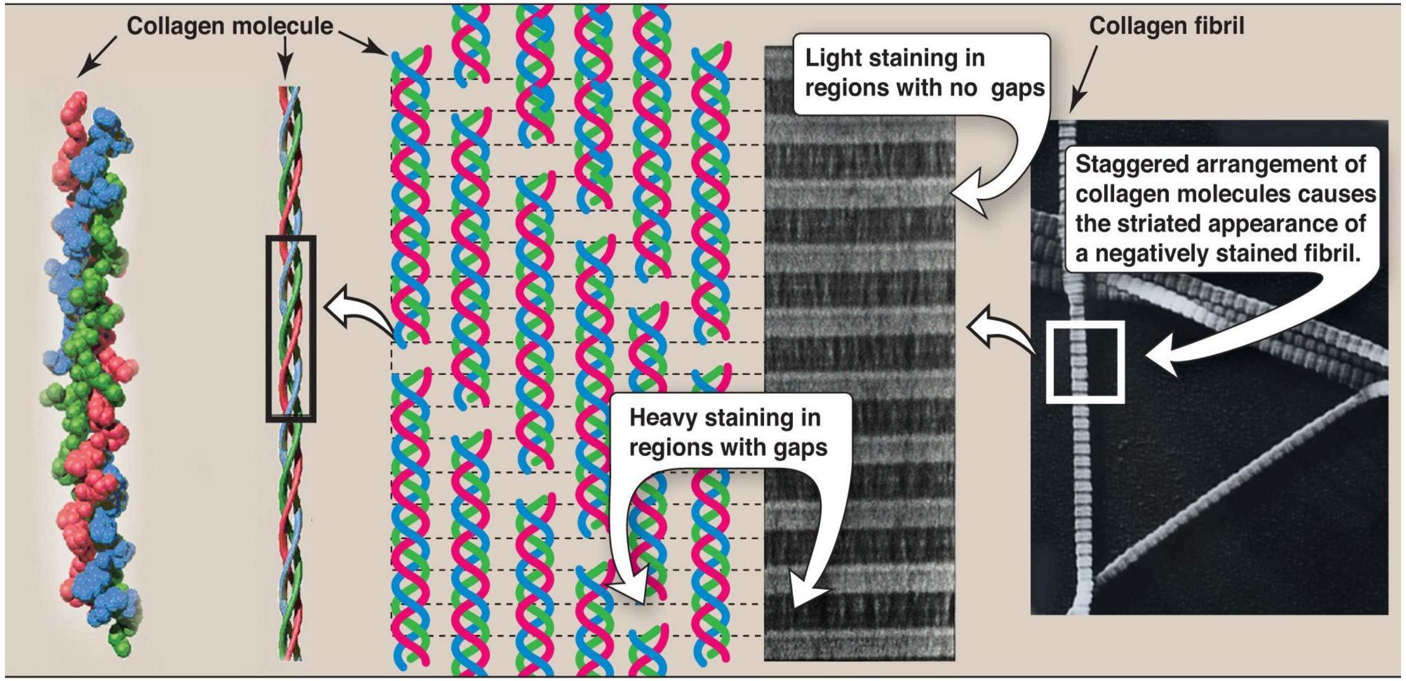
Collagen is the most abundant fibrous protein



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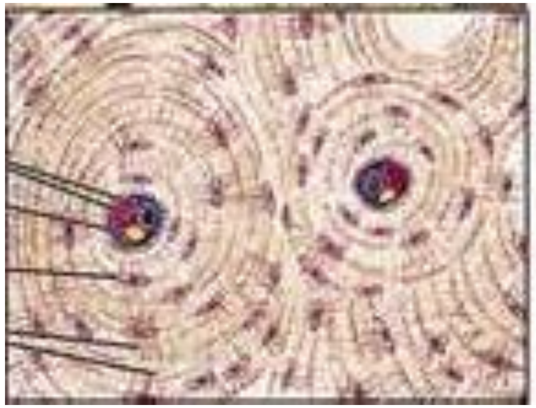
A typical **collagen molecule** is a **triple-helix** of **three α-chains** which are about 1,000 amino acids long.
This **rigid rope-like structure** can be **covalently cross-linked**.

Fibrillar collagens provide mechanical strength for examples in tendons and bones

Collagen in **tendons**
is bundled in long and cross-linked parallel fibers.

Collagen in **bones**
provides the overall structure and strength and the flexibility to resist mechanical sheer. **Calcium phosphate** is deposited in and around the collagen fiber. The collagen gaps are nucleation sites for hydroxyapatite.



Collagen has a distinctive amino acid composition

~ **Gly** – Leu – Hyp – **Gly** – Pro – Hyp – **Gly** – Ala – Hyl – ~

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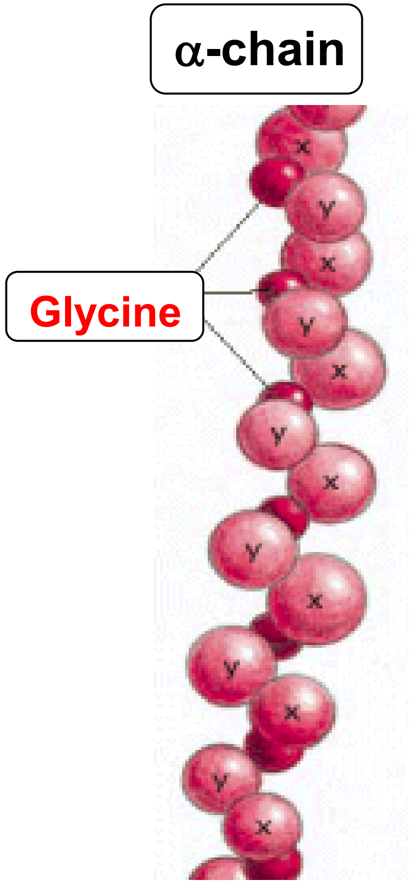
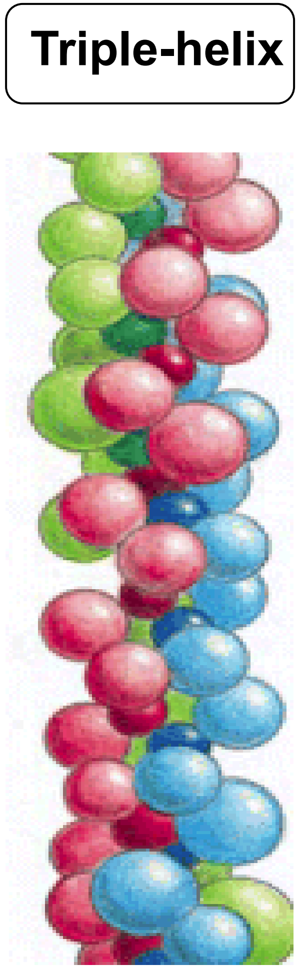


1. **Glycine** with the smallest side chain is found in each third position.

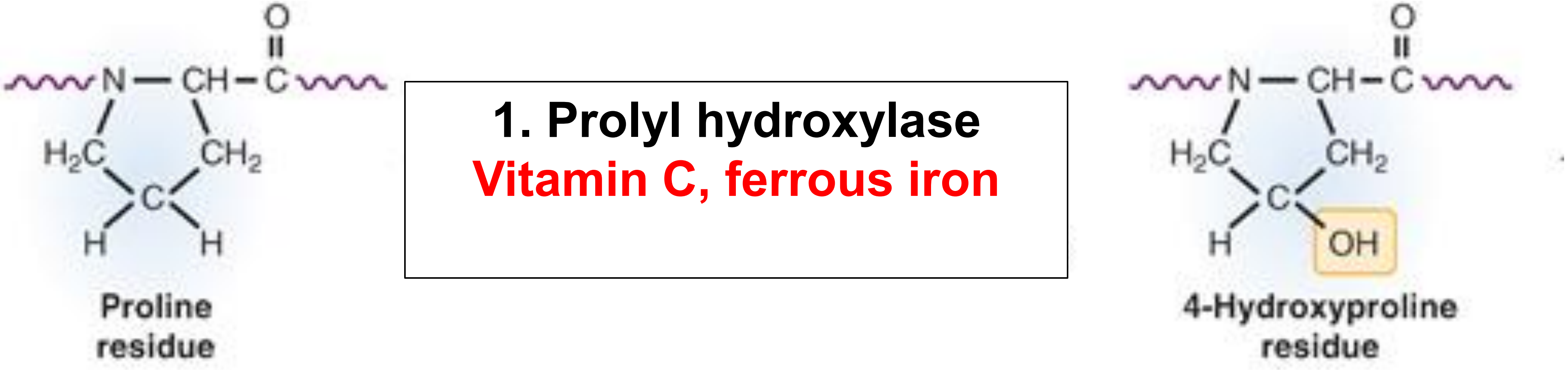
2. **Proline** and **hydroxyproline** (Hyp) residues are abundant and form “kinks” in the polypeptide chain which facilitates tight winding.

Stabilization of the collagen triple-helix



- **Hydrogen bonds** between the three **α–chains** stabilize the triple-helix.
- The bonds are formed between the peptide bonds and side chains of **glycine, proline** and **hydroxyproline.**
- Hydrogen bonds are non-covalent and it is important to involve the bonds formed with hydroxyproline residues.

Formation of Hydroxyproline residues in pro-α-chains needs vitamin C



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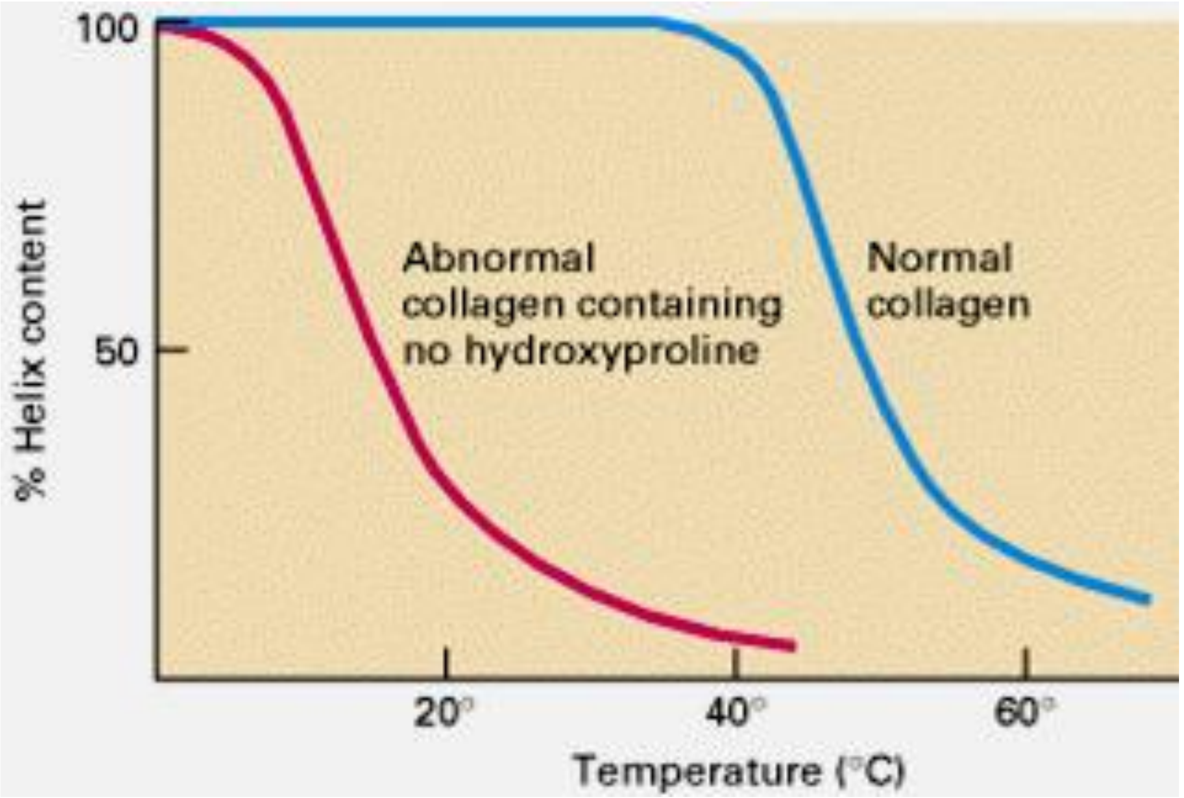
Hydroxyproline residues are needed for **collagen stability.**

Collagen without hydroxyproline residues is instable.



Lehninger

The collagen triple helix is normally intact up to 40°C.
Without hydroxyproline most of the collagen triple-helix content is lost at human body temperature.

Deficiency of vitamin C can lead to Scurvy

Decreased stability and **tensile strength** of collagen leads to: bleeding gums, hemorrhages, poor wound healing and can lead to death.

Figure 4.8
The legs of a 46-year-old man with scurvy.

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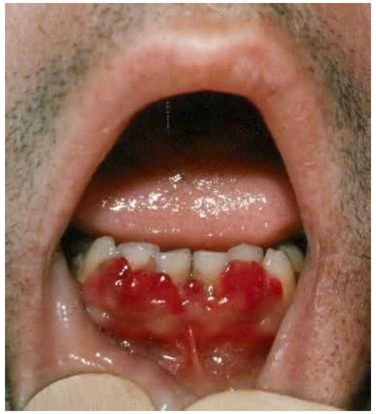

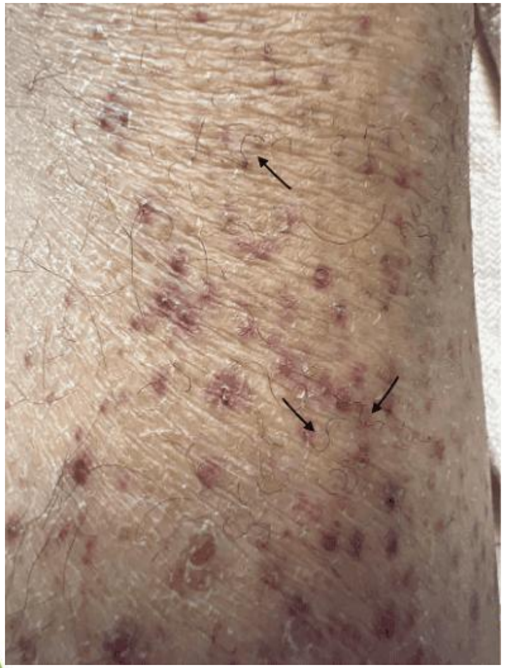

Perifollicular hemorrhages

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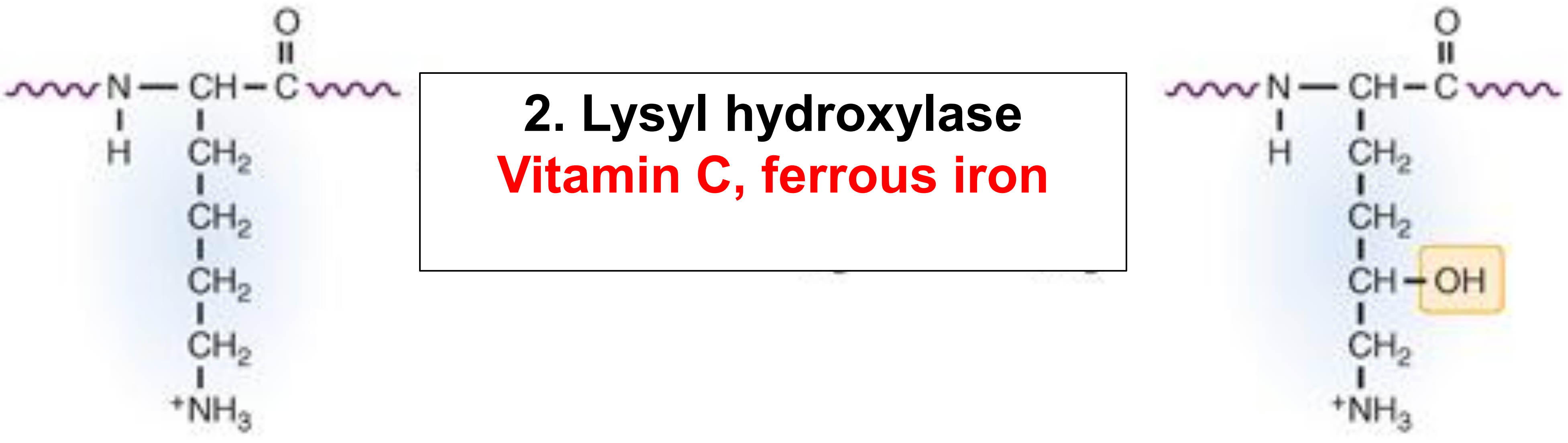
Hydroxylation of lysine residues in pro-α-chains also needs vitamin C.



Lysine residue

5-Hydroxylysine residue

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Hydroxylysine residues are needed for **O-glycosylation** leading to the large variety of collagen molecules and are also important for cross-linking by lysyl oxidase

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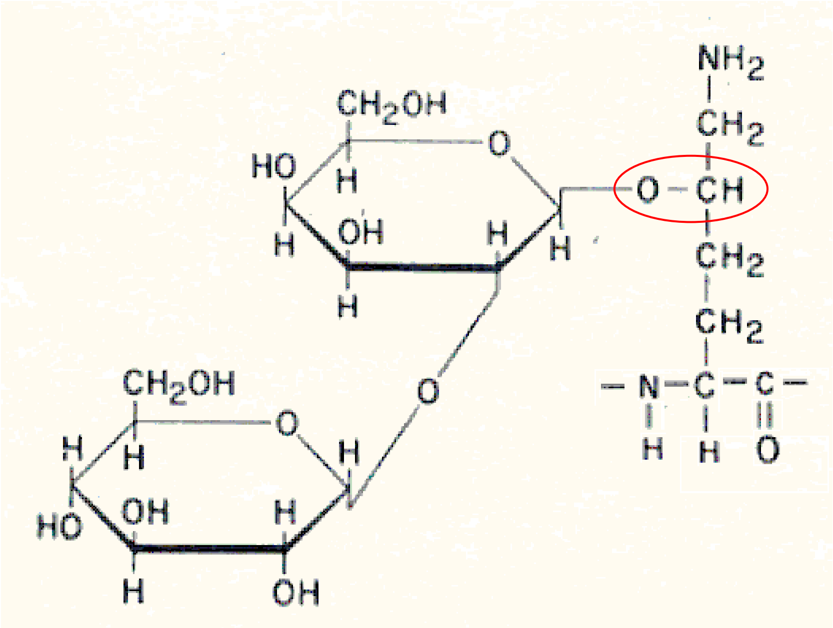
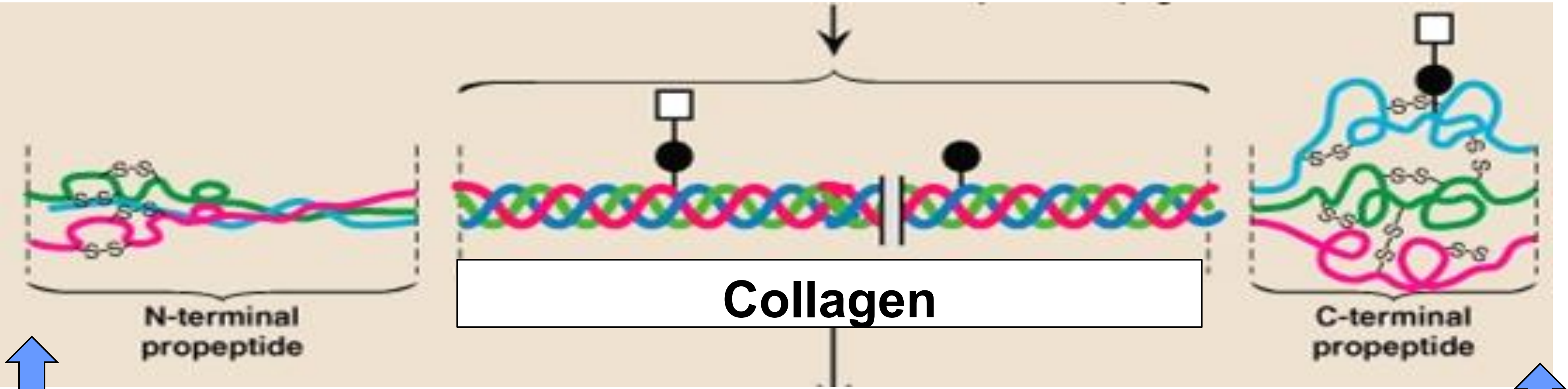
O-linked glycosylation of collagen with glucose or glucosyl-galactose



Hydroxy-lysine residues

What is special regarding collagen synthesis?

Collagen is an **insoluble protein**
and needs to be synthesized as soluble **procollagen**
before release into the ECM.



Procollagen has propeptides and is formed in fibroblasts, osteoblasts or chondroblast

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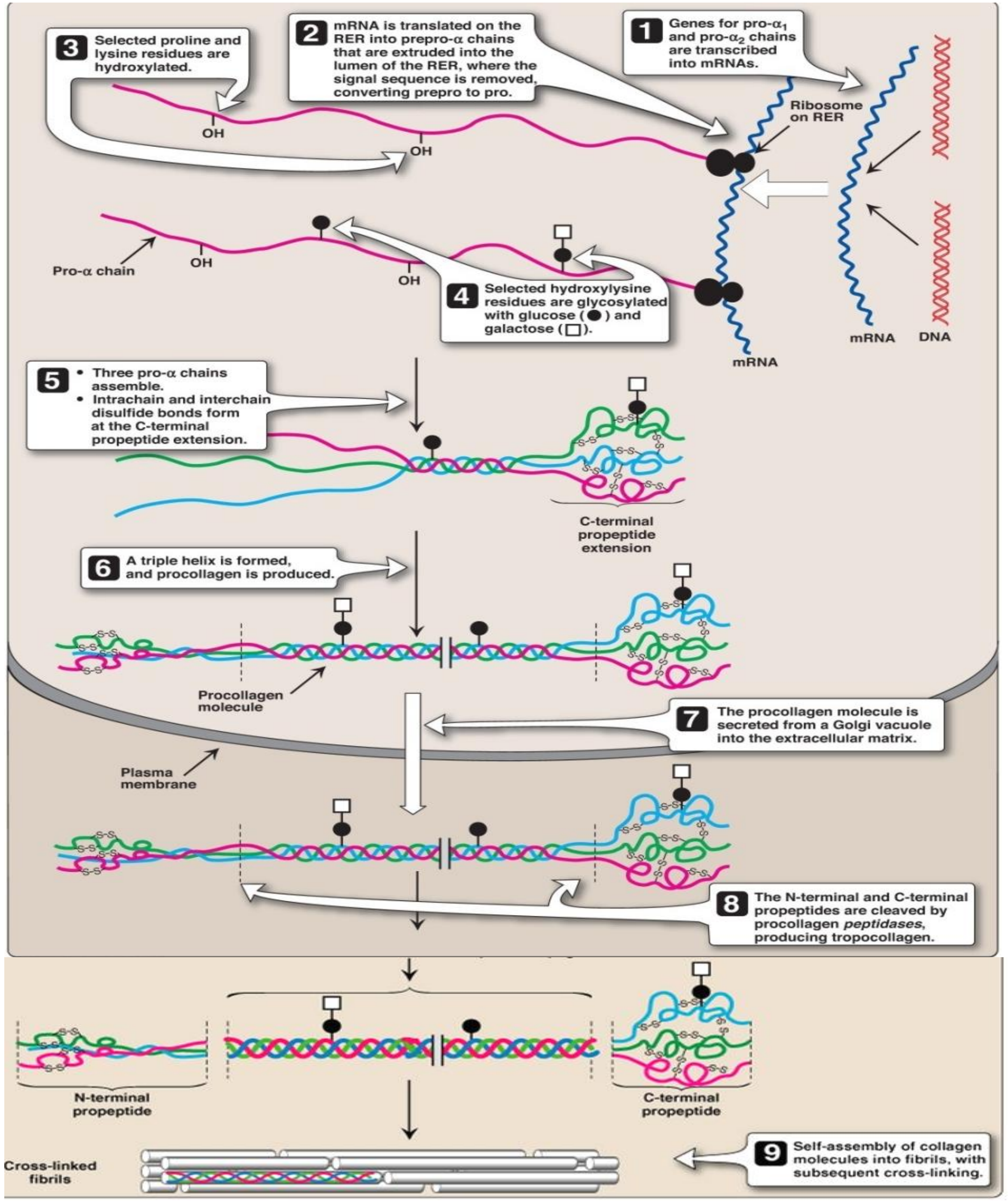
Outline of fibril-forming collagen synthesis



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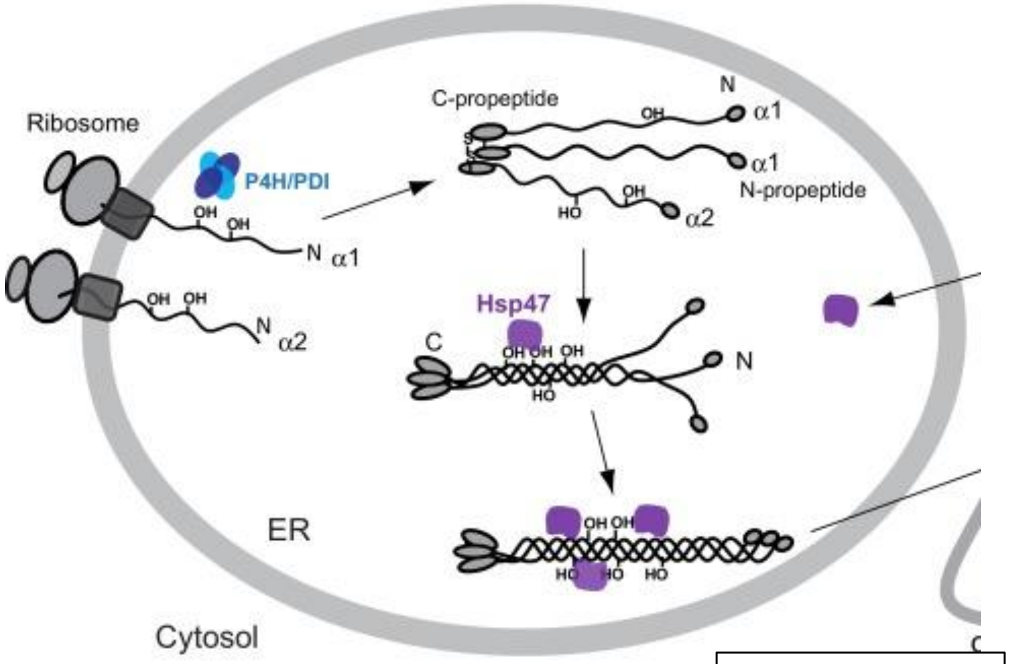
Biosynthesis of collagen Type I

Two genes are expressed: ***COL1A1*** and ***COL1A2*** which lead to **two** pro-α**1 chains** and **one** pro-α**2 chain: (**$\alpha 1_2 \alpha 2$**)** for the triple helix**.**

Chaperone or **heat shock proteins** link the correct pro-α-chains to each other by disulfide bonds at the C-terminal propeptides.



Sciencedirect.com

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Summary of collagen Type I synthesis

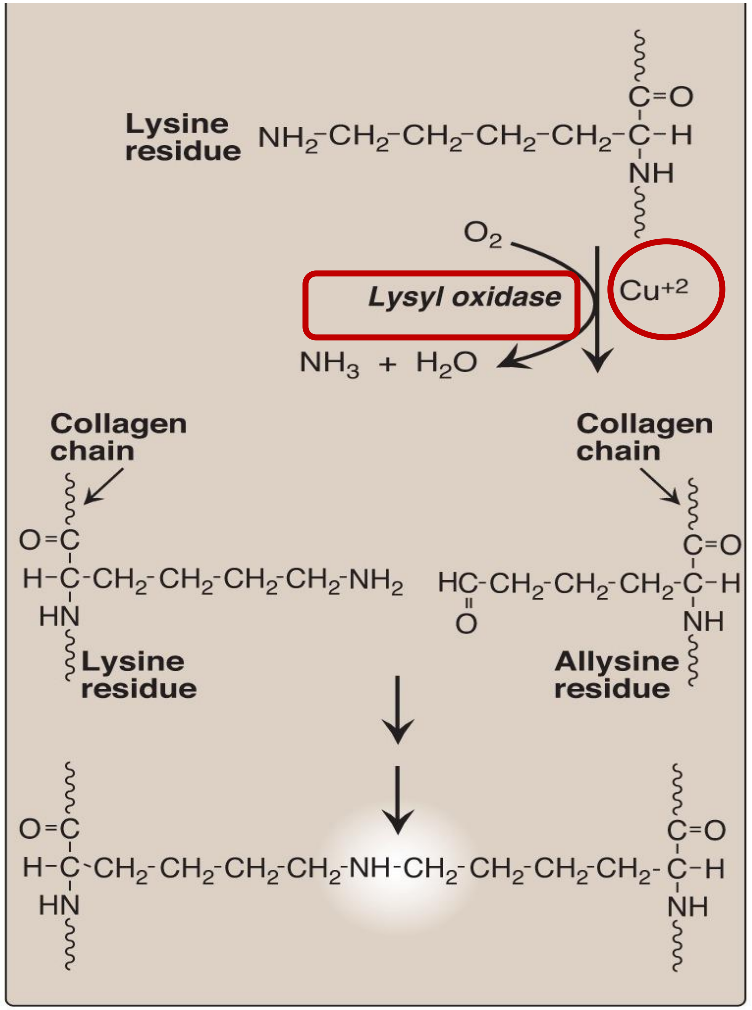
Nucleus: transcription of one or two collagen genes: for Type I collagen: *COL1A1* and *COL1A2.*

RER: 1. Synthesis of pro-α-chains into the ER lumen.
2. Hydroxylation of specific proline and lysine residues (**vitamin C**).
3. Start of O-glycosylation of selective hydroxylysine residues with glucose or galactose.
4. Chaperones: Linkage of the **3 correct α-chains** at the C-terminal by disulfide bonds **($\alpha 1_2 \alpha 2$)** and formation of the procollagen triple-helix (C- to N-terminal zipper). Closing of triple-helix with disulfide bonds at the N-terminal propeptides.

Golgi: Finishing glycosylation and procollagen is packed into secretory vesicles for release into the ECM.

ECM: Cleavage of propeptides by propeptidases leads to the insoluble collagen.
Staggered self-assembly of the collagen molecules.
Covalent extracellular cross-linking using lysyl oxidase (copper) which forms allysine and hydroxyallysine residues in collagen.

Extracellular cross-linking strengthens collagen fibers and needs lysyl oxidase (LOX) in the ECM



Lysyl oxidase oxidatively deaminates **lysine** or **hydroxylysine** residues.
LOX has a higher affinity for hydroxylysine.

Lysyl oxidase needs **copper** as cofactor and forms **aldehydes** known as **allysine** or **hydroxyallysine** residue.

The **aldehyde** is highly **reactive** and forms spontaneously a **covalent bond** with other lysine or hydroxylysine or allysine residues of another collagen molecule.

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Menkes Disease

Menke disease is an inherited X-linked disease caused by defective intestinal absorption of copper, and impaired lysyl oxidase activity.

Clinical manifestations include:

- Kinky, colorless, friable hair
- Skin hypopigmentation
- Bony abnormalities (i.e. osteoporosis)

Collagenopathies

1. **Ehlers-Danlos syndromes (EDS)**

(be aware not to mix up Collagen Types with EDS Types)

2. **Osteogenesis Imperfecta (OI)**

(be aware not to mix up Collagen Types with OI Types)

Collagenopathies

1. Ehlers-Danlos syndrome

is a **large heterogenous group** of connective tissue disorders featuring **skin hyperextensibility** and **fragility,** and **joint hypermobility** that result from abnormal synthesis or structure of **fibrillar collagen**. Symptoms can overlap and can result from:

- **Mutation** of a **collagen gene** which affects often **collagen Type V (classic EDS) or collage Type III (vascular EDS).**

- **Enzyme deficiency** of one of the **enzymes** needed for **collagen synthesis,** like for example lysyl hydroxylase or propeptidases.

The **classic form** of **EDS** (EDS Type I,II)

is mostly caused by defects in **Type V collagen** *(COL5A1, COL5A2)*.

Patients show the following:

Hyperextensibility of the skin and fragility

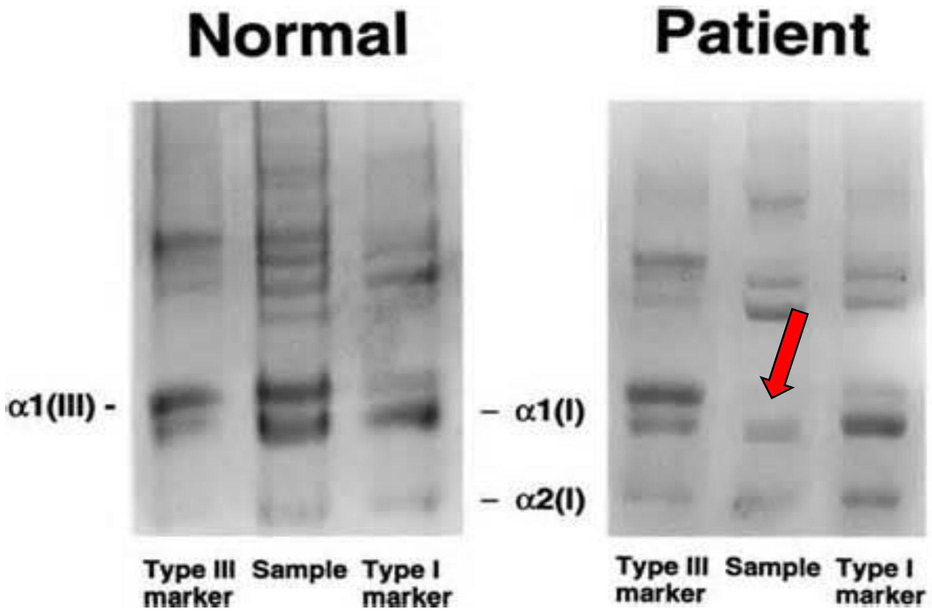
Hypermobility of the joints

The **vascular form** of **EDS** (EDS Type IV)

is caused by defects in **Type III collagen** (*COL3A1*)

and leads to **fragility** of **skin** and **vascular vessel walls.**

It is the **most severe** form as it is associated with potential **lethal rupture of large arteries and colon**.



Collagenopathies

2. Osteogenesis Imperfecta (OI)

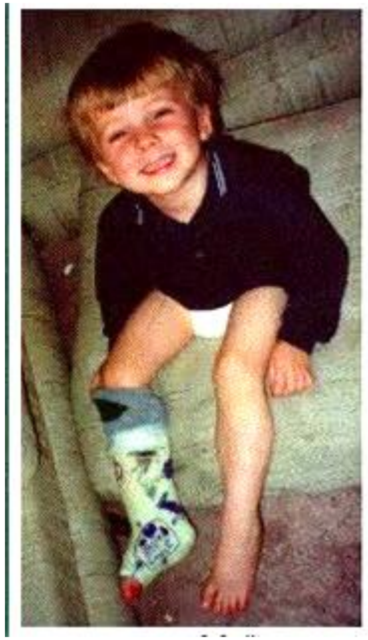
Brittle bone disease is caused in over 90% of patients by an autosomal dominant **inherited mutation** in one allele of the ***COL1A1*** or ***COL1A2*** gene and affects **type I collagen**.

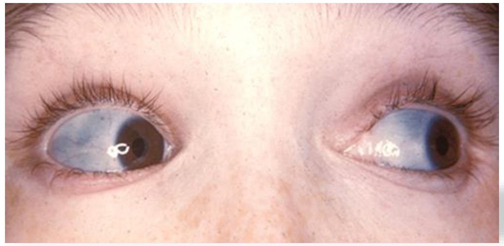
This results often in the **replacement** of **glycine** in **bone collagen** and affects the **formation** and **stability** of the **collagen triple-helix.**

OI Type I is the **mildest** form.
OI Type II is the **most severe** form.

OI Type I: mildest form and most common Osteogenesis Imperfecta Tarda

- Leads in early childhood to **long bone fractures** after minor trauma.
- In **adulthood** less **fractures.**
- Normal or near **normal height**.
- Possibility of **hearing loss** in adulthood.
- Individuals have often **blue sclerae**.
  This results from a **thin, translucent sclera** revealing the appearance of the **choroidal veins**.



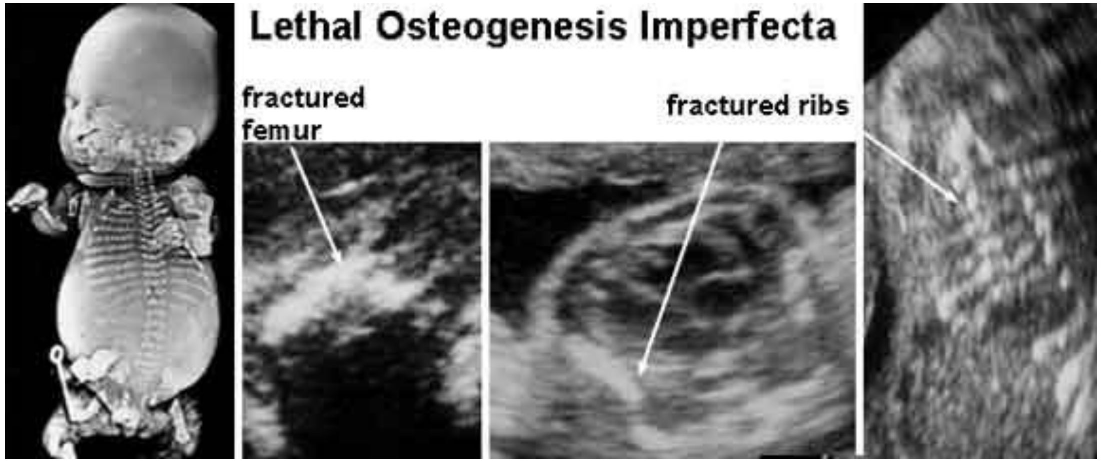


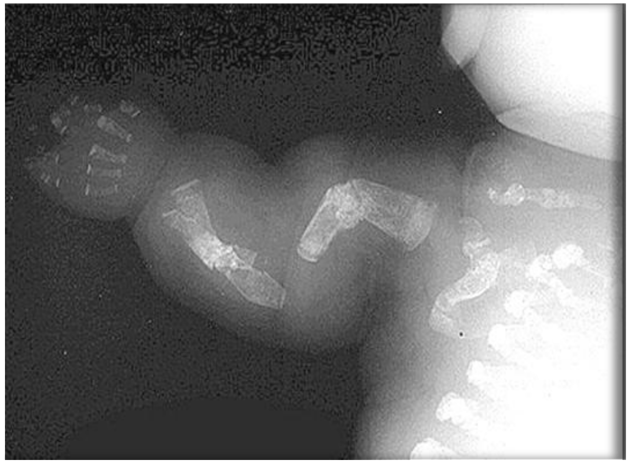
En.Wikipedia.com

OI Type II most severe form: Osteogenesis Imperfecta Congenita

- Leads to **death** in **utero** or **neonatal** death due to **respiratory problems.**

- Can include**: underdeveloped lungs** and an abnormal small **fragile rib cage.**





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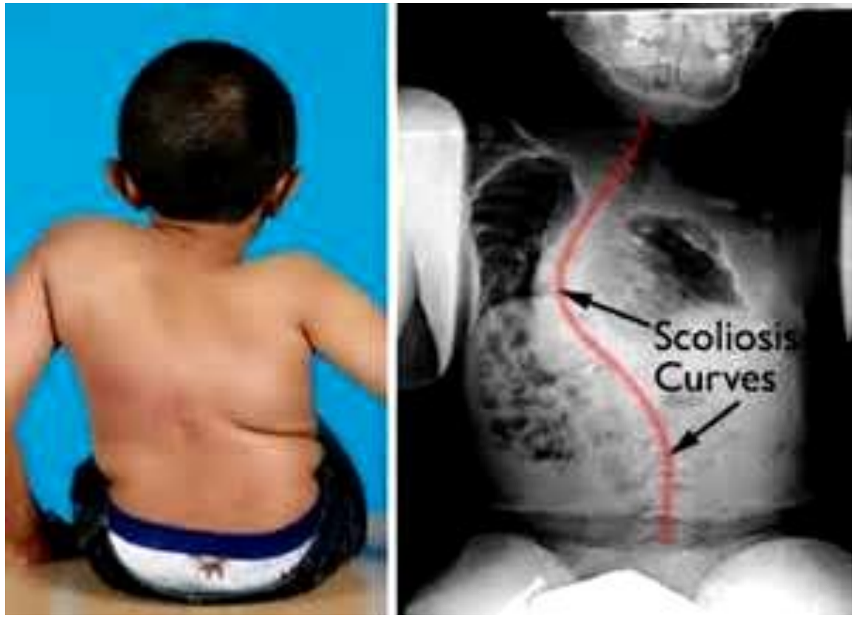
Children with Osteogenesis imperfecta Types III and IV

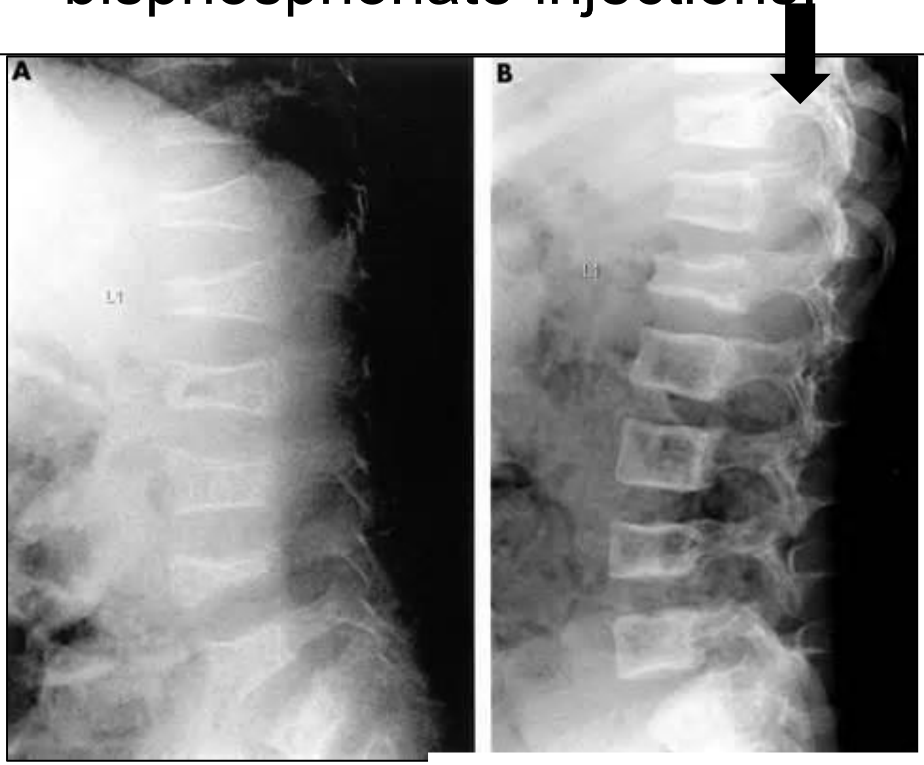
Peerj.com

Dentiogenesis imperfecta



Orthoinfo.aaos.com

A. One-year-old girl with OI Type IV
B. after 2 years of treatment with bisphosphonate injections.

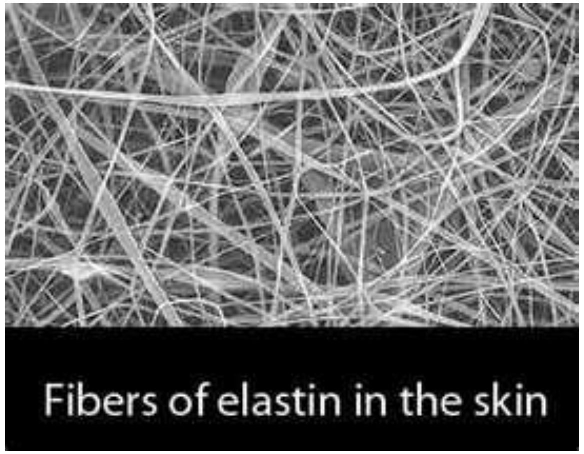


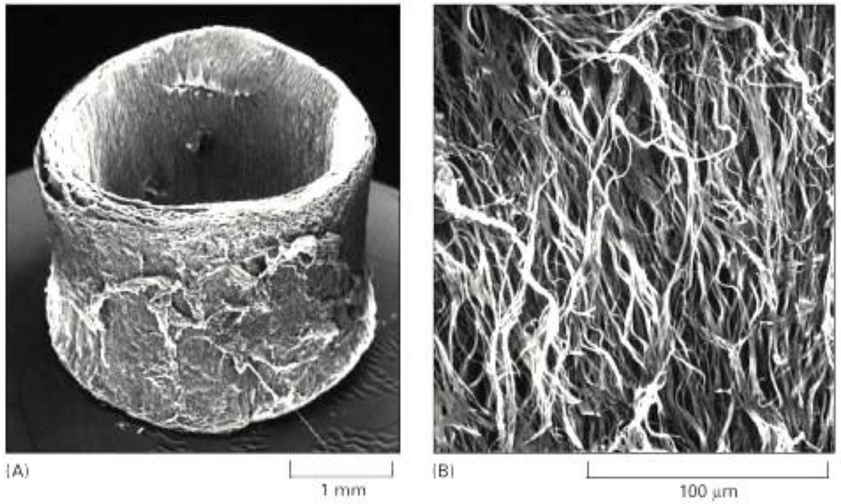
adc.bmj.com/content/86/5/356

Collagen is the most abundant fibrous protein which represents 30% of total body protein mass. It forms fibrils or mesh-like networks.

Fibrous Proteins

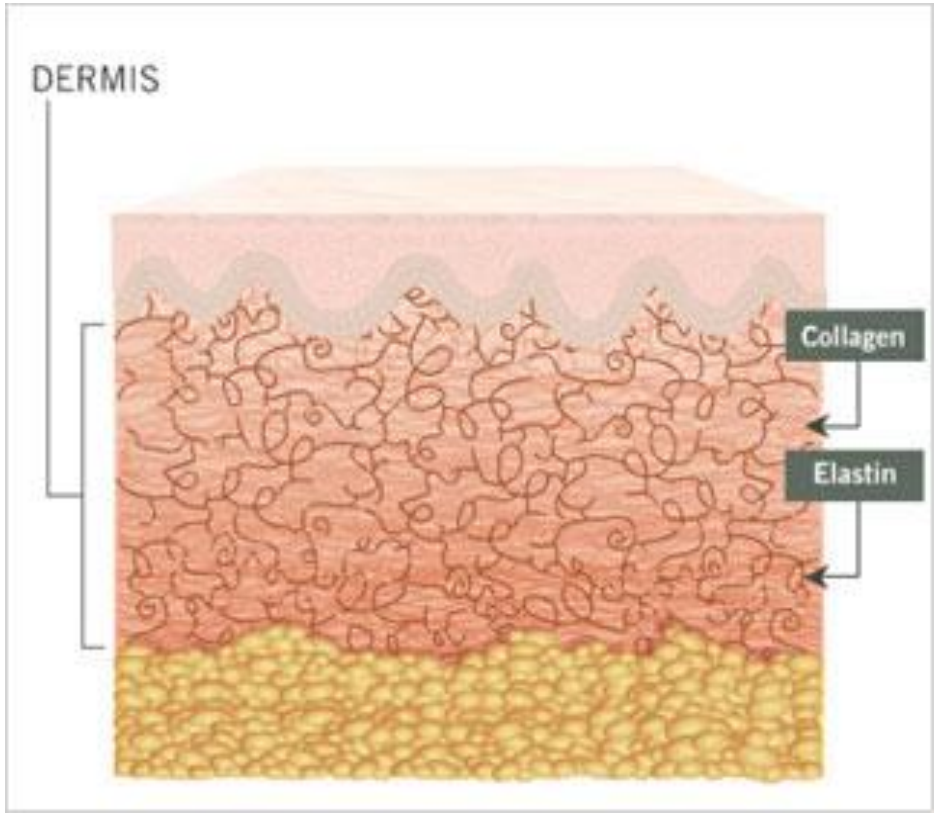
Elastin has a highly cross-linked insoluble amorphous structure. It is the **major protein** in **elastic fibers** which allow the flexibility of blood vessels, lungs, ligaments and skin.





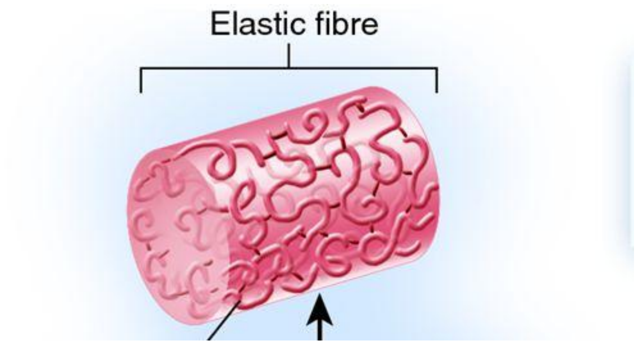
Collagen and elastin for the skin are used in cosmetics

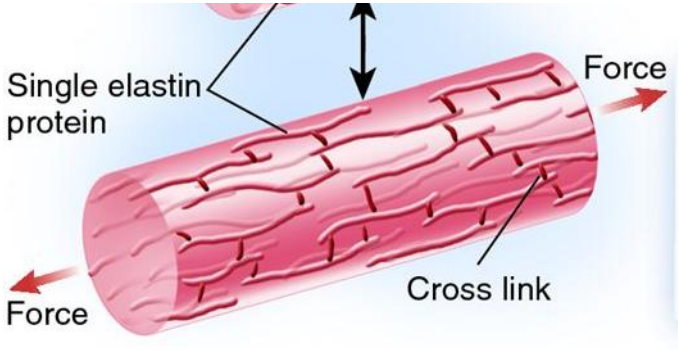
(collagen and elastin fillers)



Elastic fibers consist of elastin and microfibrils.

They can be stretched and then reform without an obvious energy source.





htpps://cellbiology.med.unsw.edu.au

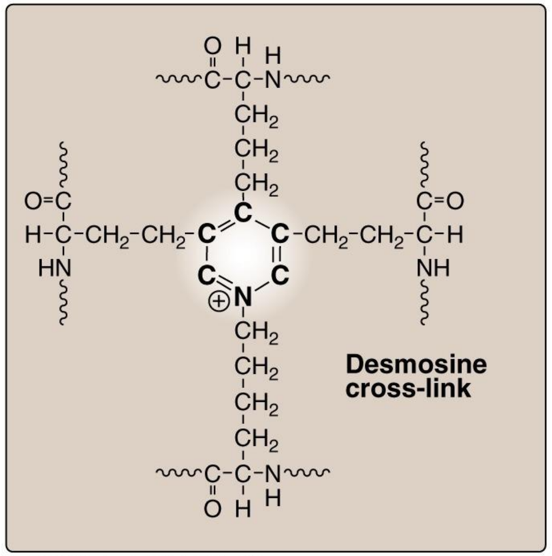
Elastin has alternating domains of:

1. **Highly hydrophobic region:** rich in **glycine, alanine, valine** and **proline.**
2. **More hydrophilic region** which contains **lysine** residues for cross-linking

The **stretched structure** of elastin exposes the **hydrophobic region** to water. When the **stretching force stops elastin reforms** to its original form.

Desmosine and **isodesmosine** are characteristic for **elastin** and allows it to **stretch** and **bend** in any direction.

Yellow color



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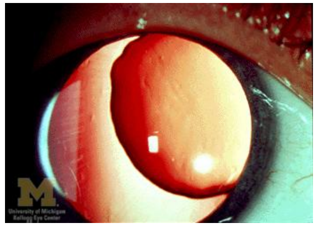
3 allysine (lysyl oxidase) and **one lysine** residue are covalently linked in **desmosine.**

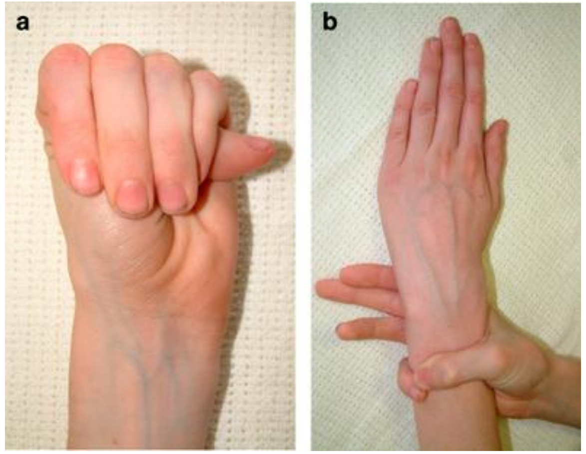
Marfan syndrome



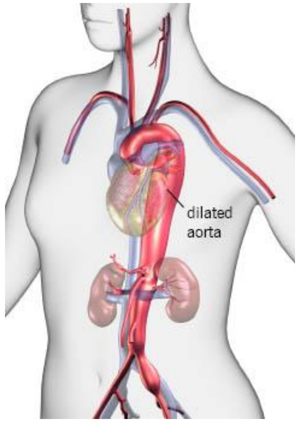
Autosomal dominant hereditary defect in the **gene** that encodes **fibrillin-1**(*FBN1*)

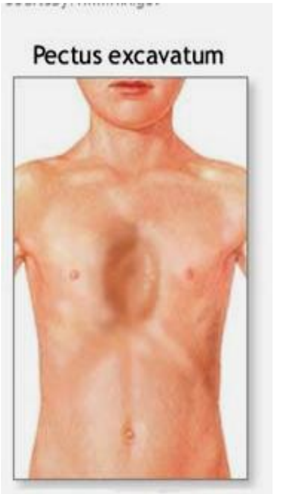
The **elastic fibers using** the **abnormal fibrillin** appear to be longer and slender and do not form a meshwork.

This leads to abnormalities of the skeleton, cardiovascular system and the eyes.
Cardiovascular lesions are the most life-threatening features.
The majority of deaths are caused by rupture of aortic dissections.



Main characteristics are:
Unusual tall, long limbs, arachnodactyly
scoliosis, increased joint flexibility,
lens dislocation
aortic root dilation
abnormal formation of rib cage





Summary and comparison related to the DLA video

O-glycosylation

Individual linkage of activated sugars step by step directly to the protein.

- The first **sugar** is linked in the **RER**. **Enzymes** bound in the **Golgi** membrane recognize the structures and **link** the correct additional **sugars** directly to the growing **glycoproteins.** The different **oligosaccharides** are often **branched.**
- The **first sugar** is linked to the **OH-group** of a **serine** or **threonine** residue of the **protein.** In **collagen** the **sugar** is linked instead to **hydroxylysine residues**.
- Used for the **glycosylation** of collagen, proteoglycans, mucins, glycoproteins and blood group substances.

N-glycosylation

Formation of a mannose-rich oligosaccharide bound to the lipid dolichol pyrophosphate.

- This **mannose-rich precursor** is linked in one step **only** to the **nitrogen** of an **asparagine** side chain.
- All **proteins** receive the same **oligosaccharide** and only later the **sugars** are individually **modified** in the **RER** and **Golgi** dependent on the **protein.**
- **Complex glycoproteins or high-mannose glycoproteins are formed in the Golgi.**
- **Golgi: Transport** for release into blood, fusion with cell membrane or transport into lysosomes.
- Transport into **lysosomes** need the **mannose 6-P** marker and if deficient, it leads to **I-cell disease.**

Following are Self-study slides

Compare and contrast	**Prolyl hydroxylase**	**Lysyl hydroxylase**	**Lysyl oxidase**
Collagen synthesis	yes	yes	yes
Elastin synthesis	**No ?, rarely**	**no**	yes
Location	**intracellular**	**intracellular**	**extracellular**
Vitamin C cofactor	yes	yes	**no**
Reduced activity in Scurvy	yes	yes	**no**
Preparation for O-glycosylation	**no**	yes	**no**
Needed for hydrogen bonds	yes	**no**	**no**
Copper cofactor	**no**	**no**	yes
Forms allysine or hydroxyallysine residues	**no**	**no**	yes
Directly needed for covalent cross-linking	**no**	**no**	Yes: collagen and elastin (desmosine)
Can be deficient in specific EDS	yes	yes	yes

Deficiencies of vitamin C and copper

Deficiency	Biochemical description	Special
Vitamin C deficiency: **Collagen synthesis:** Deficient enzymes: prolyl hydroxylase and lysyl hydroxylase	**Collagen synthesis affected:** **1.** Lack of hydroxyproline residues leads to instability of collagen triple-helix. **2.** Eventually reduced O-glycosylation due to less hydroxylysine residues in collagen.	**Vitamin C deficiency** can lead to **scurvy** with bleeding and poor wound healing. Severe form leads to death.
Copper deficiency: **Collagen and elastin synthesis:** Deficient lysyl oxidase in the ECM.	Lysyl oxidase forms allysine residues in collagen and elastin which are needed for cross-linking in collagen and in elastin and for desmosine formation in elastin.	**Copper deficiency** can lead to **aneurysms** due to less formation of desmosine in elastin. **Menkes disease** (copper intestinal absorption defect).

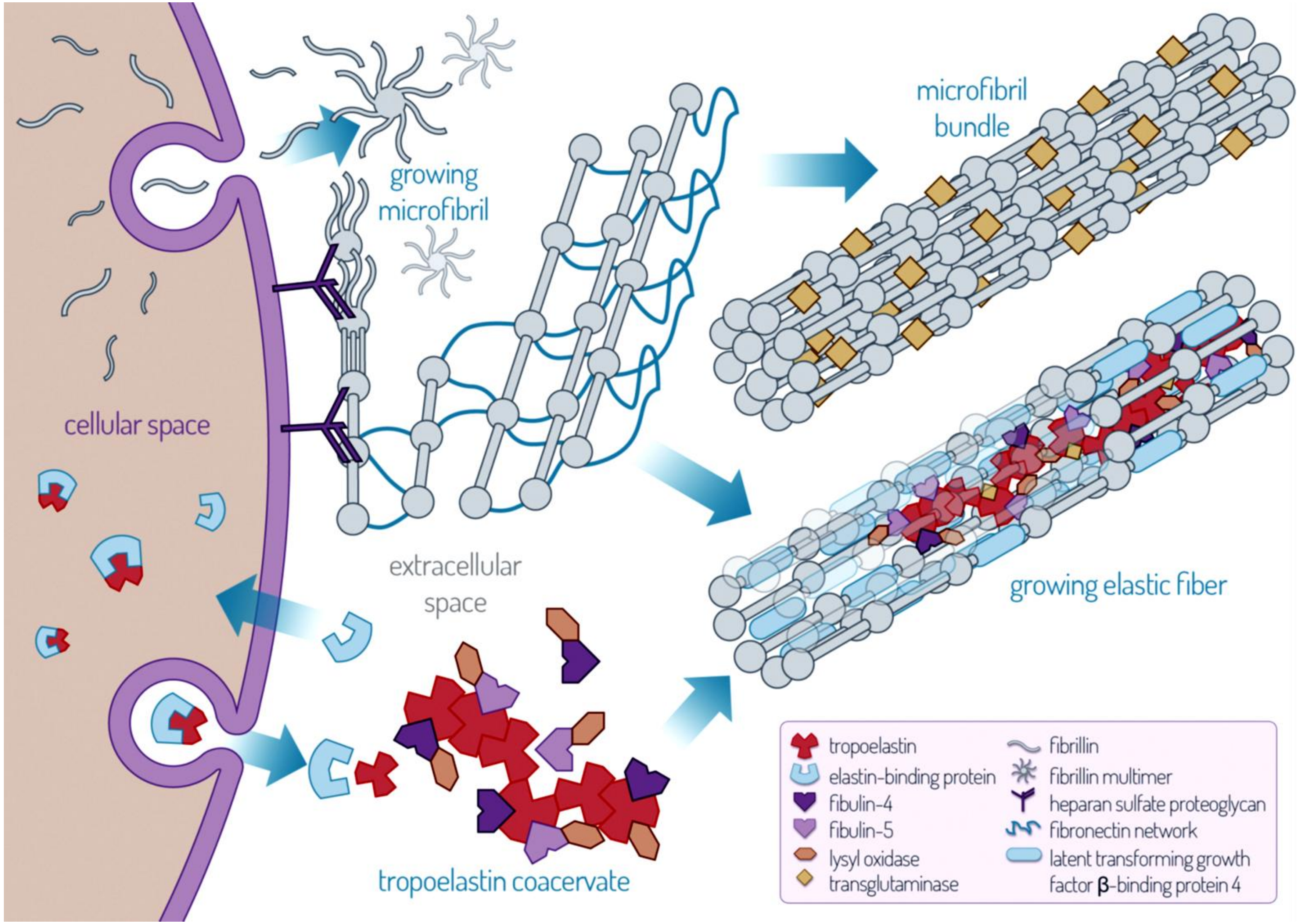
SOM.MK.I.BPM1.1.FTM.3.BCHM.0153 SOM.MK.I.BPM1.1.FTM.3.BCHM.0155

Defect or disease SOM.MK.I.BPM1.1.FTM.3.BCHM.0161	Biochemical description	Clinical features
Ehlers Danlos Syndromes: **Abnormal collagen Type III or Type V** Gene mutations or deficient enzymes needed for synthesis	**Vascular** form: (EDS IV) **collagen Type III** is defective **Classic** form: (EDS I,II) **collagen Type V** is defective	**EDS** abnormal collagen **Type III:** Fragility of skin and vascular vessel walls, can be lethal. **EDS** abnormal collagen **Type V:** Hypermobility of joints and hyperextensibility of skin
Osteogenesis Imperfecta: **Brittle bone disease** **Abnormal collagen Type I**	**Collagen Type I** is defective: Often a mutation of *COL1A1* or *COL1A2* related to a glycine residue leading to an instable collagen triple-helix.	**OI Type I:** mildest, blue sclerae. **OI Type II**: most severe, perinatal death. **OI Types III and IV:** severe with scoliosis, bone malformations and dentiogenesis imperfecta.
Marfan syndrome **Abnormal elastic fibers**	**Mutation in *fibrillin-1* gene:** elastin lacks its scaffold. Abnormal fibrillin leads to abnormal elastic fibers.	Very tall, long limbs, arachnodactyly, lens dislocation, **aortic root dilation**, abnormal rib cage.

Comparison	Collagen	Elastin
Amino acid composition	1/3 represented **by glycine** (in each **3rd position**	1/3 represented by glycine but found in hydrophilic, hydrophobic regions and not in 3rd position like in collagen.
Commonly found amino acids	**Glycine, proline, hydroxyproline, lysine, hydroxylysine.**	**Glycine, alanine, valine, proline, lysine rarely hydroxylated.**
Post-translational hydroxylations, vitamin C	Yes: prolyl hydroxylase and lysyl hydroxylase, vitamin C needed.	No: Elastin is mostly normal in vitamin C deficiency.
Glycosylations	Yes: some **hydroxylysyl** residues can be in glycosylated. (no glycosylation of hydroxyprolyl residues)	No
Synthesis in fibroblasts as soluble protein	Yes: procollagen triple-helix, 3 pro-α-chains. Procollagen becomes insoluble outside the cell after cleavage of propeptides by propeptidases.	Yes: **linear** tropoelastin. Needs scaffold of fibrillin to allow cross-linking and desmosine formation to become insoluble.
Genes	One or two genes are expressed for triple-helix formation.	One gene, specific splicing.
Special structure	Triple-helix of collagen stabilized by **hydrogen bonds** (hydroxyproline important).	Needs the extracellular protein **fibrillin** as scaffold.
Lysyl oxidase, copper needed	Yes: **allysine** and **hydroxy-allysine residues** for covalent cross-linking.	Yes: allysine residues and special: **desmosine** formation in elastin.
Special	**Collagen Type I-III** fibril-forming. **Collagen Type IV** for basement membrane.	Alternating **hydrophobic** and **hydrophilic** domains, stretch with energy and recoil by itself.

Comparison of defects in	Collagen	Elastin
Hereditary defects	**Ehlers-Danlos syndromes** **Collagen** Type III abnormal: **vascular form**, blood vessels. **Collagen** Type V abnormal: **classic form**, hypermobility and hyperextensibility. **Osteogenesis Imperfecta** **OI Type I:** least severe, long bone fractures in childhood, blue sclerae in some patients. **OI Type II:** most severe, death in utero or shortly after birth. **OI Type III** and **Type IV:** severe, bone malformations, teeth.	**Related: Marfan's syndrome**: deficiency of fibrillin-1 gene which leads to abnormal elastic fibers. Elastin gene itself is normal but elastin cannot be efficiently cross-linked and the abnormal fibrillin is found in elastic fibers.
Cofactor defects	**Vitamin C** deficiency: Scurvy Deficient prolyl hydroxylase (instability) and lysyl hydroxylase (less O-glycosylation). **Copper** deficiency: lysyl oxidase deficient, less cross-linking between collagen strands.	**Copper** deficiency: lysyl oxidase is deficient and less cross-linking and desmosine formation can lead to aneurysms.

If you want to know: How is an elastic fiber synthesized?



Text for previous slide: **Synthesis of elastic fiber**

- The **elastic fiber** is formed from an insoluble **elastin core (90%)** and a peripheral mantle of **fibrillin-rich microfibrils**.

- There is only **one human elastin gene** but different splicing of **mRNA** leads to different **isoforms. Tropoelastin** is secreted from **elastogenic cells** into the **ECM** as a highly **soluble linear polypeptide** (about 700 amino acids) which aggregates at the cell surface**.**

- **Cross-linking** is initiated by **lysyl oxidase** and the premature elastin clusters detach from the cell surface once they reach a critical size. Proteins guide them to the microfibrillar scaffold where they align and are further cross-linked using lysyl oxidase which forms allysine residues resulting in desmosine and isodesmosine which is only found in elastin.

- The microfibrillar scaffold consists mainly of the large glycoprotein **fibrillin-1** which not only lends biomechanical support but also communicates with **integrins** and **controls growth factor signaling**.

Thank you, if you have questions please email me.

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